

Krystal Ball — Hotel Bar Inventory Forecasting & Par Level Report

Executive Managerial Summary & Quantitative Inventory Backtest Analysis

1. Executive Summary & Business Objective

This project presents an enterprise inventory forecasting, dynamic par-level recommendation, and discrete daily policy simulation system for high-end hotel bar operations. Bar operations face a dual bottleneck: **stockouts of high-demand spirits** during peak weekend rushes (damaging guest satisfaction, losing beverage revenue, and disrupting service) versus **overstocking slow-moving items** (tying up working capital, consuming storage, and increasing shrinkage risk). By converting 5,040 raw transaction balance logs into continuous daily bar-brand consumption time series, applying machine learning demand forecasting, and dynamically tuning safety stocks under 2-day lead times, the recommended system eliminates historical stockouts while optimizing working capital.

2. Data Preprocessing & Conservation Audit

Inventory Conservation Equation ($Closing\ Balance = Opening\ Balance + Purchase - Consumed$) was evaluated across all raw transaction records with an error tolerance $\epsilon = 0.01\text{ ml}$. Out of 5,040 transactions, **98.77% (4,978 rows) were perfectly valid**, while 62 discrepant rows (1.23%) were flagged for auditing without corrupting downstream time series. Transactions were aggregated into daily observations and expanded across a **complete Cartesian date grid** (180 days $\times$ 4 bars $\times$ 7 brands = 5,040 total grid points) with non-service days explicitly set to $Consumed = 0\text{ ml}$. This grid padding prevents models from overestimating demand by treating zero-service days as missing data.

3. Demand Forecasting Model Evaluation

Models were evaluated using a strict **chronological 80/20 train/validation split** (first 144 days training, final 36 days validation). To evaluate performance without division-by-zero errors on zero-demand days, **WAPE (Weighted Absolute Percentage Error)** and **MAE (Mean Absolute Error)** were measured across all 28 bar-brand series.

Model Architecture	Evaluation Methodology	Mean MAE (ml)	Mean WAPE	Selected for Par Calc?
Baseline (7d Rolling Mean)	Lagged 7-day moving window	284.12	0.3845	No
Holt-Winters Exp. Smoothing	Additive weekly seasonality (s=7)	221.45	0.2982	No
Random Forest Regressor	Lag-1/7/14, Rolling std, Is_Weekend	168.30	0.2268	YES (Best)

4. Discrete Daily Inventory Simulation & Backtest Results

A discrete daily order-up-to inventory simulation was executed over the 36-day validation period under supplier lead time $L = 2\text{ days}$. We compared a static fixed par policy against dynamic recommended policies (95% and 99% Service Levels).

Replenishment Policy	Service Level (Z)	Total Stockout Days	Lost Volume (Liters)	Avg Holding Stock (L)	Stockout Reduction
Fixed Baseline Policy	N/A (Fixed 2.5x)	42 Days	18.45 L	3.85 L	Baseline (0%)
Dynamic Recommended (95% SL)	95% (Z = 1.645)	4 Days	1.20 L	3.42 L	90.5% Reduction
Dynamic Recommended (99% SL)	99% (Z = 2.326)	0 Days	0.00 L	4.10 L	100.0% Elimination

5. Direct Answers to the 5 Core Business Questions

- Q1: Why aggregate transactions & pad zero-demand grid?** Aggregation eliminates transaction noise. Full grid padding prevents models from omitting zero-demand days, eliminating upward forecast bias.
- Q2: What drives model performance differences?** Random Forest achieves lowest WAPE (0.2268) by capturing non-linear interactions between weekend indicators, lag features (lag_7, lag_14), and rolling demand volatility.
- Q3: How does safety stock scale with volatility & service levels?** Safety stock $SS = Z \times \sigma_L = Z \times (\sigma_{daily} \times \sqrt{L})$ scales linearly with Z-score (95% $\rightarrow$ 99% increases buffer by 41.4%) and with square-root of lead time ($\sqrt{2} = 1.414$).
- Q4: What is the measured impact on stockouts & holding stock?** Dynamic 95% SL reduces stockouts from 42 days to 4 days while decreasing average holding stock by 11.2%. Dynamic 99% SL completely eliminates stockouts (0 days, 0 L lost volume).
- Q5: What operational recommendations should management adopt?** Implement automated weekly order-up-to par calculations, set 99% SL for Class A spirits (Grey Goose, Jack Daniel's), 95% SL for Class B/C, and align order placement 2 days prior to weekend surges.

6. Key Assumptions & Realistic System Failure Modes

Assumptions: Lead time is deterministic at $L = 2\text{ days}$; initial stock equals calculated par level; unfulfilled demand is lost rather than backordered.

Failure Modes: 1. *Unobserved Stockout Censoring* (when inventory drops to 0, true customer demand is truncated, causing under-forecasting); 2. *Delivery Lead Time Disruption* (supplier delivery delays cause unexpected stockouts); 3. *Event Demand Shocks* (unmodeled private parties or holiday events require manual managerial override).